Instruction manual

An ISO CERTIFIED COMPANY

MARS

Mars EdPal Instruments Pvt. Ltd.

TECHNICAL SUPPORT : For Complaints/Suggestion, Please call our Customer Care number at 09215880005
or Send the Email on : sales@marsedpal.com

PORTAL FRAME APPARATUS

INSTRUCTION MANUAL

MARS

Mars EdPal Instruments Pvt. Ltd.

AMBALA, (HARYANA)

PORTAL FRAME APPARATUS

1.0 APPARATUS REQUIRED:

- Portal frame Apparatus
- Dial Gauge : 1 No.
- Magnetic Base : 1 No.
- Hangers : 4 Nos.
- 100 gm weights : 5 Nos.
- 200 gm weights : 5 Nos.
- 500 gm weights : 5 Nos.

2.0 THEORY:

Structures are categorized as statically determinate or as statically indeterminate. Determinate structures can be analyzed by using the conditions of equilibrium, where as indeterminate structures needs additional conditions to solve. The portal frame is an indeterminate structure to several degree of indeterminacy depending on the end conditions.

To know the behavior of any frame it is advisable to know its different deflected shapes under different loading conditions, which can be obtained by vertical work energy method analytically.

Portal structures similar to the end portals of bridge have as their primary purpose, the transfer of horizontal loads applied at their top to their foundations. Clearance requirements usually lead to the use of statically indeterminate structural layout for portals, and approximate solutions are often used in their analysis.

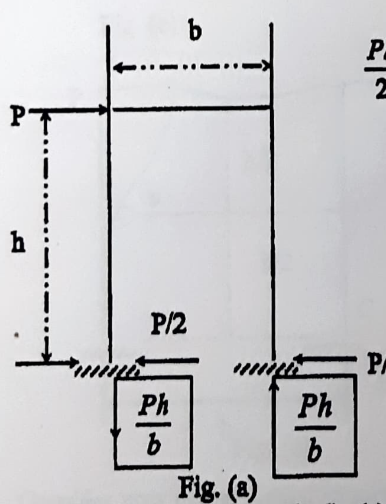


Fig. (a)

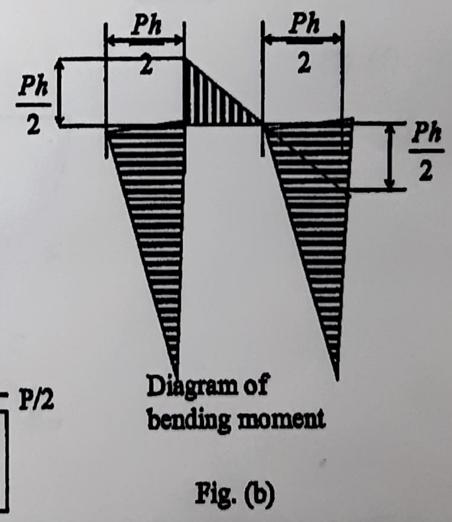


Fig. (b)

Consider the portal shown in fig (a) all the members of which are capable of carrying bending and shear as well as axial force. The legs are hinged at their base and rigidly connected to the cross girder at the top. This structure is statically indeterminate to the first degree; hence, one assumption must be made. Solutions of this type of structure, based on elastic considerations, show that the total horizontal shear on the portal will be divided almost equally between the two legs; it will therefore be assumed that the horizontal reactions for the two legs are equal to each other and therefore equal to $P/2^2$. The remainder of the analysis can now be carried out by statics. The vertical reaction on the right leg can be obtained by taking moments about the hinge at the base of the left leg. The vertical reaction on the left leg can then be found by applying $\Sigma f_y = 0$ to the entire structure. Once the reactions are known, the diagrams of bending moment and shear are easily computed, leading to values for bending moment as given in fig (b). It is well to visualize the deformed shape of the portal under the action of the applied load.

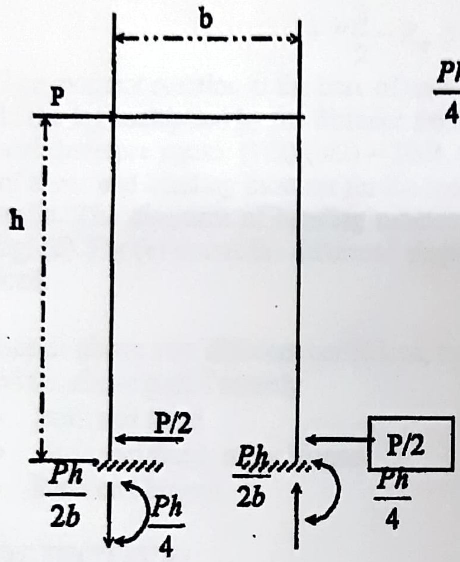


Fig. (c)

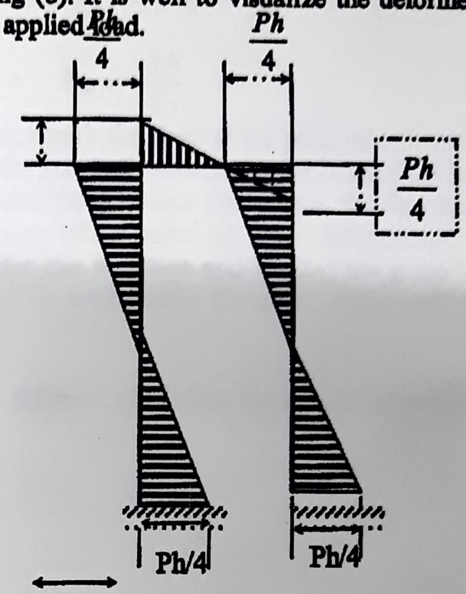


Fig. (d)

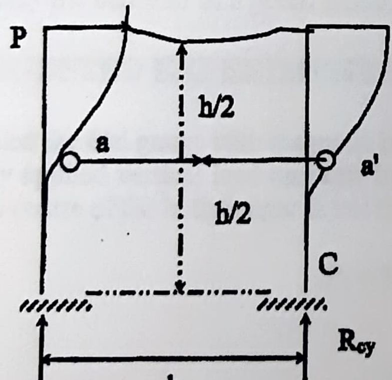


Fig. (e)

Consider now portal similar in some ways to that of fig. (a) but with the bases of the legs fixed, as shown in fig (c). This structure is statically indeterminate to the third degree, so that three assumptions must be made. As when the legs were hinged at their base, it will again be assumed that the horizontal reactions for the two legs are equal and hence equal to P/2. It will be noted that near the center of each leg there is point of reversal of curvature. These are points of inflection, where the bending moment is changing sign and hence has zero value. It will therefore be assumed that there is a point of inflection at the center of each leg; this is structurally equivalent to assuming that hinges exist at points a and a', as shown in fig (e). The vertical reactions on this portal equal the axial forces in the portal legs and can be determined by successively taking moments about a and a' of all the forces acting on that portion of the structure above a and a'.

For example, taking moments about a gives

$$+P\frac{h}{2}-R_{ey}\,b=o \qquad R_{ey}=+\frac{Ph}{2b}$$

The moment reaction at the base of each leg equals the shear at the point of inflection in the leg multiplied by the distance from the point of inflection to the base of the leg and therefore equals (P/2) (h/2) = Ph/4. Once the reactions are known, the diagrams of shear and bending moment for the members of the portal are easily determined by static. The diagrams of bending moment for this structure and loading are given in fig. (d). Fig (e) shows the deformed shape of the portal under the action of the applied load.

Beside above two different conditions, two different conditions can also be visualized on the above portal namely

- Both end fixed
- Once end fixed, other hinged.
- Both end hinged.

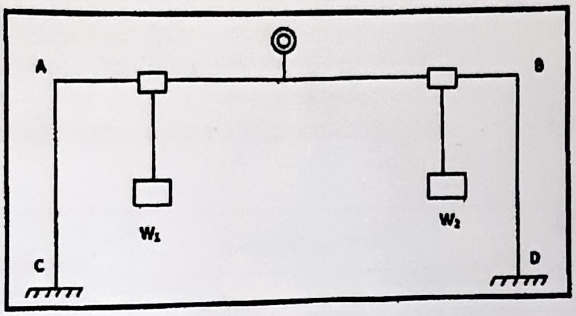
3.0 OBJECTIVE:

To study the behavior of a portal frame under different end conditions.

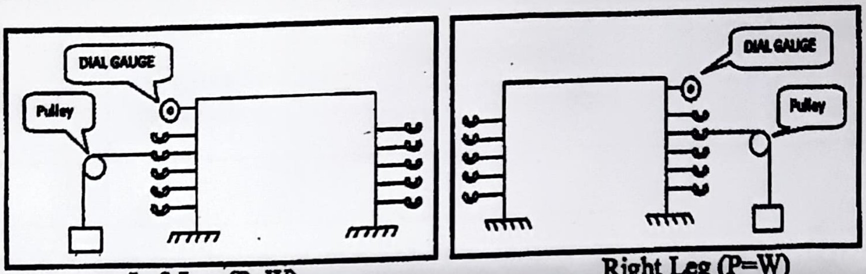
4.0 SUGGESTED EXPERIMENTAL WORK:

1. Installed the dial gauge with magnetic bas & wire both left and right side.
2. Firstly applied vertical load on same length or different side hanger and set the dial gauge centre of the both hanger & not the deflection.

$$P_1 = W_1, P_2 = W_2$$



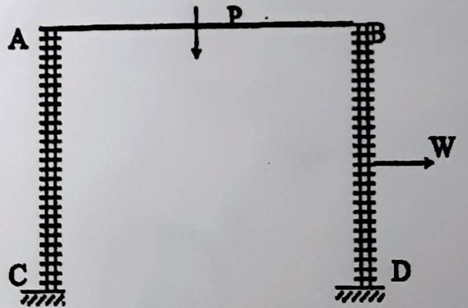
3. Another condition is select the leg left or right side and applied the weight at various points and measures the defection through dial gauges.



Left Leg (P=W) Right Leg (P=W)

4. Provide accessories like roller, hinged and fixed condition.
5. Change the condition as per your choice.
6. Condition is Fixed-Fixed, Fixed-Roller.
7. Unload the frame and shift the dial gauge to another leg and repeat the above (2) & (3).
8. Tabulate the observed reading and sketch the deflected shape for the portal frame on the graph sheet.
9. Repeat the all steps for various ends conditions and loading conditions to obtain the deflected shape.

5.0 SAMPLE DATA SHEET:



Portal frame with end conditions = both fixed/both hinged/fixed-hinged

Loading point x on AC /AB/ BD =
Loading Point AB each Cut distance = 5 cm
Beam Thickness = 6 mm
AC/BD Point hooks distance measure from top of the plane = 7 cm each
Load applied = ...kg

Points on AC	Distance of point from C	Dial gauge reading		Deflections (mm)
		Initial	Final	

Points on AB	Distance of point from A	Dial gauge reading		Deflections (mm)
		Initial	Final	

Points on BD	Distance of point from D	Dial gauge reading		Deflections (mm)
		Initial	Final	

6.0 COMMENTS:

MARS